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PRESSURE VESSEL BOSS FOR A ROTO-MOLDED LINER

Abstract

A boss is configured for attachment to a liner of a pressure vessel. The boss includes a port, a neck and a flange. The port has a longitudinal axis and connects an interior and exterior of the pressure vessel. The flange includes an outer edge at a farthest radial extent from the longitudinal axis; an exterior side on a first side of the outer edge; and an interior side on an opposed second side of the outer edge. The interior side includes a first radially inner annular groove and a second radially outer annular groove. Each of the first and second annular grooves has a width at its opening that is equal to or greater than a width at its base. An assembly includes a liner having a first opening and a first boss attached to the liner at the first opening.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims the benefit of priority from U.S. Provisional Application No. 63/556,454, filed Feb. 22, 2024. This application also claims the benefit of priority from U.S. Provisional Application No. 63/750,332, filed Jan. 28, 2025. The priority applications are hereby incorporated by reference in their entireties.

BACKGROUND
[0002] Pressure vessels are commonly used for containing a variety of fluids under pressure, such as storing oxygen, natural gas, nitrogen, propane and other fuels, for example. Suitable container materials include laminated layers of wound fiberglass filaments or other synthetic filaments bonded together by a thermal-setting or thermoplastic resin. An elastomeric or other non-metal resilient liner or bladder often is disposed within the composite shell to seal the vessel and prevent internal fluids from contacting the composite material. The composite construction of the vessels provides numerous advantages such as lightness in weight and resistance to corrosion, fatigue and catastrophic failure. These attributes are primarily due to the high specific strengths of the reinforcing fibers or filaments that are typically oriented in the direction of the principal forces in the construction of the pressure vessels.
[0003] FIGS. **1** and **2** illustrate an elongated pressure vessel **10**, such as that disclosed in U.S. Pat. No. 5,476,189, which is hereby incorporated by reference. Vessel **10** has a main body section **12** with end sections **14**. A boss **16** is provided at one or both ends of the vessel **10** to provide a port for communicating with the interior of the vessel **10**. The vessel **10** is formed from an inner polymer liner **20** covered by an outer composite shell **18**. In this disclosure, “composite” means a fiber reinforced resin matrix material, such as a filament wound or laminated structure.
[0004] The liner **20** has a generally hemispheroidal end section **22** with an opening **24** aligned within an opening **26** in the outer composite shell **18**. Boss **16** is positioned within the aligned openings and includes a neck portion **28** and a radially outwardly projecting flange portion **30**. The boss **16** defines a port **32** through which fluid at high pressure may be communicated with the interior of the pressure vessel **10**.
[0005] Liner **20** includes a dual layer lip circumscribing opening **24** in the liner **20**, with an exterior lip segment **34** and an interior lip segment **36** defining an annular recess **38** therebetween for receiving flange portion **30** of boss **16**. Dovetailed inter-engaging trap locks **40** are provided between flange portion **30** and exterior and interior lip segments **34** and **36**, respectively, to lock liner **20** to boss **16**.
[0006] This type of interlocking liner and boss structure is quite effective when the liner is formed by injection molding. However, the structure is not ideal for liners that are formed in a rotational molding process, which results in a liner having relatively uniform thickness. One prior roto-molded tank seal is disclosed in U.S. Pat. No. 7,837,054 by Van Oyen, assigned to Quantum Fuel Systems Technologies Worldwide, Inc. of Irvine, California. The Van Oyen tank seal uses a multiple piece assembly that holds a radial seal (o-ring) **63** between a liner **61** and a boss member **64**, resulting in complexity in assembly and use.
SUMMARY
[0007] In one aspect, a boss is configured for attachment to a liner of a pressure vessel. The boss comprises a port, a neck and a flange. The port has a longitudinal axis and is configured to connect an interior of the pressure vessel and an exterior of the pressure vessel. The neck circumscribes an exterior portion of the port. The flange circumscribes an interior portion of the port and extends

radially outward from the neck. The flange comprises an outer edge at a farthest radial extent from the longitudinal axis; an exterior side disposed on a first side of the outer edge; and an interior side disposed on an opposed second side of the outer edge. The interior side comprises a first radially inner annular groove and a second radially outer annular groove positioned at a greater distance from the longitudinal axis than the first radially inner annular groove. The first radially inner annular groove has a first width at its first base and a second width at its first opening, wherein the second width is equal to or greater than the first width. The second radially outer annular groove has a third width at its second base and a fourth width at its second opening, wherein the fourth width is equal to or greater than the third width. In another aspect, an assembly comprises a liner comprising a first opening and first boss attached to the liner at the first opening.

[0008] This summary and the Abstract are provided to introduce concepts in simplified form that are further described below in the Detailed Description. This summary is not intended to identify key features or essential features of the disclosed or claimed subject matter and is not intended to describe each disclosed embodiment or every implementation of the disclosed or claimed subject matter. Specifically, features disclosed herein with respect to one embodiment may be equally applicable to another. Further, this summary is not intended to be used as an aid in determining the scope of the claimed subject matter. Many other novel advantages, features, and relationships will become apparent as this description proceeds. The figures and the description that follow more particularly exemplify illustrative embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The disclosed subject matter will be further explained with reference to the attached figures, wherein like and analogous structure or system elements are referred to by like reference numerals, or numbers indexed by 100, throughout the several views. All descriptions are applicable to like and analogous structures throughout the several embodiments, unless otherwise specified.

[0010] FIG. **1** is a side elevation view of a typical elongated pressure vessel.

[0011] FIG. **2** is a partial cross-sectional view through one end of such a pressure vessel, taken along line **2-2** of FIG. **1**.

[0012] FIG. **3** is a cross-sectional perspective view of a conventional boss.

[0013] FIG. **4** is a cross-sectional perspective view of a first embodiment of a boss of the disclosure.

[0014] FIG. **5** is a partial cross-sectional perspective view of the first embodiment of the boss, as viewed from the right side of FIG. **4**.

[0015] FIG. **6** is a partial cross-sectional view through half of one end of a pressure vessel liner formed by roto-molding with the first embodiment of the boss.

[0016] FIG. **7** is similar to FIG. **6** but shows a second embodiment of a boss of the disclosure.

[0017] FIG. **8** is a cross-sectional perspective view of a pressure vessel having the second embodiment of a boss disposed at both opposed ends.

[0018] FIG. **9** is an enlarged view of the encircled portion “**9**” of FIG. **8**.

[0019] FIG. **10** is a partial cross-sectional view of the sealing portion of a third embodiment of a boss of the disclosure.

[0020] FIG. **11** is an enlarged view of the encircled portion “**11**” of FIG. **10**.

[0021] FIG. **12** is an interior end view of an exemplary boss of the disclosure, having six relatively narrow slots for a liner anchor.

[0022] FIG. **13** is an interior end view of an exemplary boss of the disclosure, having eight relatively narrow slots for a liner anchor.

[0023] FIG. **14** is an interior end view of an exemplary boss of the disclosure, having eight relatively wider slots for a liner anchor.

[0024] FIG. **15** is a partial cross-sectional view through half of one end of a pressure vessel liner formed by roto molding, illustrating radial shrinkage of the liner material about the exemplary pressure vessel boss.

[0025] While the above-identified figures set forth one or more embodiments of the disclosed subject matter, other embodiments are also contemplated, as noted in the disclosure. In all cases, this disclosure presents the disclosed subject matter by way of representation and not limitation. It should be understood that numerous other modifications and embodiments can be devised by those skilled in the art that fall within the scope of the principles of this disclosure.

[0026] The figures may not be drawn to scale. In particular, some features may be enlarged relative to other features for clarity. Moreover, where terms such as above, below, over, under, top, bottom, side, right, left, vertical, horizontal, etc., are used, it is to be understood that they are used only for ease of understanding the description. It is contemplated that structures may be oriented otherwise.

[0027] The terminology used herein is for the purpose of describing embodiments, and the terminology is not intended to be limiting. Unless indicated otherwise, ordinal numbers (e.g., first, second, third, etc.) are used to distinguish or identify different elements or steps in a group of elements or steps and do not supply a serial or numerical limitation on the elements or steps of the embodiments thereof. For example, “first,” “second,” and “third” elements or steps need not necessarily appear in that order, and the embodiments thereof need not necessarily be limited to three elements or steps. Unless indicated otherwise, any labels such as “left,” “right,” “front,” “back,” “top,” “bottom,” “forward,” “reverse,” “clockwise,” “counter clockwise,” “up,” “down,” or other similar terms such as “upper,” “lower,” “aft,” “fore,” “vertical,” “horizontal,” “proximal,” “distal,” “intermediate” and the like are used for convenience and are not intended to imply, for example, any particular fixed location, orientation, or direction. Instead, such labels are used to reflect, for example, relative location, orientation, or directions. The singular forms of “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

DETAILED DESCRIPTION

[0028] Referring to FIGS. **2** and **3**, when a liner **20** is formed by injection molding, molten polymer material can be made to flow around the structures of boss **16**, such as flange **30**, to fill in the recesses therein to form exterior lip segment **34**, interior lip segment **36**, and trap locks **40**. As shown in FIG. **2**, the thickness of liner **20** around these various features is not uniform. For example, thicker deposits of liner material make up the trap locks **40** that fill grooves **41**, as well as the area **38** at the radially distal end of flange **30**; other areas of liner **20** are thinner, such as at exterior lip segment **34**, interior lip segment **36**, and main body section **12**. Another boss and liner interface is described in commonly owned U.S. Pat. No. 9,644,790 by Newhouse for “Pressure Vessel Boss and Liner Interface,” which is hereby incorporated by reference.

[0029] A liner can also be formed by a rotational molding (also referred to as roto-molding) process that uses a roto-molding machine. An exemplary roto-molding liner manufacturing process involves introducing powder resin into a mold cavity of the machine and heating the powder to a specific temperature to liquefy it. The mold is then rotated to coat the resin in a layer on all sides of the interior of the mold and a boss attached to the mold. This layer is allowed to cure to result in a hollow resin structure. In a case in which a multiple layer structure of a liner is desired, the process can be repeated for each layer. Because formation of a liner by roto-molding builds up the liner by coating the mold cavity with the molten resin, the resulting liner generally has a uniform thickness around the inside of the mold and around structures of the boss around which the liner is formed. This disclosure describes exemplary embodiments of a pressure vessel boss **116** that is particularly configured for effective sealing in a pressure vessel **10** having a liner **120** formed by rotational molding.

[0030] FIGS. **4-15** illustrate exemplary embodiments of a boss **116** configured for use with a roto-molded liner **120**. Three specific embodiments of a pressure vessel boss **116** are described, and in some cases they will be differentiated by referring to the first embodiment with reference number **116**_a_ (FIGS. **4-6**), the second embodiment with reference to number **116**_b_ (FIGS. **7-9**), and the third embodiment with reference to number **116**_c_ (FIGS. **10** and **11**). However, in many aspects, the structures are similar; descriptions of boss **116**, **116**_a_, **116**_b_ or **116**_c_ apply to all embodiments unless otherwise specified. This convention also applies to other similarly numbered elements.

[0031] Referring to FIG. **8**, in an exemplary embodiment, a liner **120** and its associated boss **116** together form a portion of a pressure vessel **110** having an interior environment **50** and an exterior environment **51**. Structures such as surfaces designated as “interior” or “exterior” will have orientations relative to the interior environment **50** and the exterior environment **51**. For example, flange **130** has an interior surface **60** and an exterior surface **62** (labeled in FIG. **9** for example).

[0032] While only portions of a boss **116**, liner **120** and shell **18** of pressure vessel **110** are illustrated in some figures, so that their structures are more clearly visible, it is to be understood that each of the boss **116**, liner **120**, shell **18** and associated pressure vessel structures is annular in shape, with half of the structures shown in FIGS. **4**, **8** and **9**, for example. In exemplary embodiments, the boss **116** is rotationally symmetrical, so that the unillustrated portion is a mirror image of the illustrated half. In an exemplary method for forming liner **120** in a roto-molding machine, the boss **116** is placed in the molding chamber so that edge **68** contacts the mold cavity surface, and the liner material does not form on the exterior surface **62** of flange **130**.

[0033] In an exemplary embodiment, as shown in FIGS. **4-6**, in a first exemplary embodiment of a boss **116**_a_, ridge **54** extends interiorly of flange **130** and provides a structure on and through which liner **120**_a_ is secured to boss **116**_a_. In an exemplary embodiment, annular groove **44** is provided on boss **116**_a_ between flange **130** and interior surface **56**, into which the liner material may flow to form an interlocking bead **45** of liner **120** (labeled in FIG. **6**). Groove **44** extends generally radially outwardly from axis **42** of boss **116** and of a pressure vessel **110** formed thereon. In an exemplary embodiment, ridge **54** includes radially inner annular lip **58** extending in a radial direction inwardly to prevent separation of boss **116** and liner **120** in the axial direction (along axis **42**). In an exemplary embodiment, a generally annular groove **46** is disposed between interior surface **60** of flange **130** of boss **116** and ridge **54**, and disposed radially outwardly from groove **44**. In an exemplary embodiment, radially outer annular lip **52** extends in a radial direction outwardly from annular groove **46**, also to prevent separation of boss **116** and liner **120** in the axial direction (along axis **42**).

[0034] Referring to FIG. **6**, during a rotational molding process, liner material flows into the groove **44** and the groove **46**, and around their respective annular lips **58** and **52** to form interlocking beads **45** and **47** (on the boss surfaces and within grooves **44** and **46** of boss **116**) to prevent separation of boss **116** and liner **120** in the axial direction. Referring to FIG. **5**, in an exemplary embodiment, the radially extending lips **52** and **58** of ridge **54** are connected by bridges **70** between apertures **48**. In an exemplary embodiment, the interior surface **60** of flange **130** approaches groove **46** at about a 45 degree angle. In an exemplary embodiment, groove **46** is smoothly radiused; in the illustrated examples of bosses **116**_a_ and **116**_b_, groove **46** undercuts the lip **52** so that bead **47** is locked in place both radially and axially.

[0035] FIGS. **4** and **5** are partial cross-sectional perspective views of a first embodiment of an exemplary boss **116** without the liner formed thereon. As shown in FIG. **5**, in an exemplary embodiment, a plurality of discrete apertures **48** extend through interior surface **56** to connect to groove **44** radially outward of inner lip **58**. Though apertures **48** are illustrated as substantially oval in shape, they may be circular, polygonal, or have any other shape. In addition, though apertures **48** are illustrated as circularly disposed about annular lip **58**, they maybe otherwise located to connect groove **44** and interior surface **56**.

[0036] FIGS. **12**, **13** and **14** show interior end views of embodiments of an exemplary boss **116** having different variations in the configurations of apertures **48**. For example, FIG. **12** shows six relatively narrow apertures **48**. In exemplary embodiments, the plurality of apertures **48** are evenly spaced about the circumference of ridge **54**, but they need not be so arranged. FIG. **13** shows eight relatively narrow apertures **48**. FIG. **14** shows eight relatively wide apertures **48**. While only a few different configurations of apertures are illustrated herein, it is to be understood that they can be provided in numbers and shapes other than illustrated. The sizes, locations and shapes of apertures **48** determine the structure of anchors **49** of liner material formed therein. Thus, larger (by volume) apertures **48** lead to more robust anchors **49**; this consideration should be balanced with the strength of ridge **54**, which is maximized with smaller apertures **48**.

[0037] Referring to FIGS. **4**, **6-9** and **15**, in exemplary embodiments, flange **130** is shaped so that the interior surface **60** and exterior surface **62** taper together and meet at a perimeter edge **68** that contacts an interior surface of the molding chamber of the roto-molding machine. Accordingly, liner material smoothly coats the interior surface **60** of the flange **130** and continues (radially outwardly relative to axis **42**) onto the interior of the mold cavity. A roto-molding method of forming liner **120** includes allowing a fluid polymer material for liner **120** to coat the interior surfaces of boss **116**, including interior flange surface **60**, groove **46**, around lip **52**, on interior surface **56**, into and around apertures **48**, around lip **58** and groove **44**. Each layer of liner material cures and solidifies on those surfaces of boss **116** and the interior surfaces of the mold chamber, as the liner **120** and boss **116** assembly are joined together by such a roto-molding process.

[0038] In some embodiments, additional layers of liner material are roto-molded onto the previously cured layers to build up a liner **120** that has substantially uniform thickness around all the mold and internal boss surfaces. In FIG. **6**, a total thickness T of a completed liner **120** is shown. In an exemplary embodiment, a dimension of each of grooves **44** and **46** in the axial direction (along axis **42**) ranges from substantially equal to T to preferably no greater than twice T. Moreover, in an embodiment, dimension B in FIG. **6**, a radially extending distance (normal to axis **42** between lip **58** and the bore surface **32**) is substantially equal to T. Typically, a radially inner surface (bottom of dimension B as illustrated) is molded against a mandrel that holds the boss **116** in a roto-molding machine. Preferably, B is no smaller than T as illustrated; the area between lip **58** and a mandrel, valve or plug that would be inserted into port **32** should not fill with liner material before groove **44** is completely filled with liner material. Additionally, as shown in FIG. **5**, in an exemplary embodiment a width of the aperture **48**, designated as dimension A (in a radial dimension relative to axis **42**), is no greater than twice the thickness T, so that the liner material fully fills the aperture **48**, forming anchor **49** (see FIG. **9**) of liner material. Sizing the structures of boss **116** in this manner prevents the formation of empty pockets around the boss **116** that are void of liner material.

[0039] Each of the grooves **44***a, b, c* and **46***a, b, c* is configured for successful rotational molding, being formed with a width at groove base D that is less than or equal to a width at groove opening E. In exemplary embodiments, each groove is smoothly curved at its closed base. In this disclosure, the width at groove base D is considered to be a primary base width outside of the radiused end. The width at opening E is taken in a plane parallel to the width at groove base D. The width measurement planes are considered to be substantially parallel to the layer orientations of roto-molded layers of liner material that are deposited onto the closed bases of each of the grooves **44***a, b, c* and **46***a, b, c*.

[0040] Referring to FIG. **6**, in boss **116***a* at radially inner groove **44***a*, the width E at the groove opening is substantially equal to the width D at the groove base. Referring to FIG. **7**, in boss **116***b* at radially inner groove **44***b*, the width E at the groove opening is greater than the width D at the groove base, increasing along a radial length of the groove **44***b* at a linear rate. Referring to FIG. **10**, in boss **116***c* at radially inner groove **44***c*, the width E at the groove opening is greater than the width D at the groove base, increasing along a radial length of the groove **44***b* at a non-linear rate (floor **64** has an inclined outer radial portion **138** and a straight inner radial portion **136**). Other groove variations are possible, preferably with no area in which a width near the groove opening is narrower than a width near the groove base, to prevent the formation of liner voids during rotational molding.

[0041] In contrast, in the prior art boss **16** of FIGS. **2** and **3**, the dovetail joint configuration of each of grooves **41** has a groove base width D that is greater than the groove opening width E. Thus, when a liner **20** is roto-molded onto such a boss **16**, layers of liner material are likely to build up and close the smaller groove opening at E, likely leaving a void in the larger groove **41**, thereby creating undesirable discontinuities in the liner **20**.

[0042] This potential problem is avoided with the disclosed bosses **116**, in which grooves **44**, **46** continuously increase in dimension from base width D toward opening width E (or at least do not narrow in dimension from the groove base toward the groove opening). When liner **120** is fully formed, the liner material fills in groove **44**, groove **46** and apertures **48**, thereby creating annular beads **45**, **47** and anchors **49**, respectively. Liner **120** is thus mechanically interlocked with boss **116** by anchors **49** formed within apertures **48**, connecting the liner material on interior surface **56** with the liner material in groove **44** (bead **45**). Accordingly, even under extreme pressure conditions, separation of liner **120** from boss **116** is prevented. In effect, the liner material **120** would tear apart before separation of liner **120** and boss **116** could occur.

[0043] In an exemplary embodiment, each anchor **49** contacts only the interior side of the boss **116** (such as to the right of edge **68** in FIGS. **6** and **7**) and does not contact the exterior side of the boss **116** (such as to the left of edge **68** in FIGS. **6** and **7**). Because groove **44**, groove **46** and apertures **48** are all located on the interior side of boss **116**, there is no passage for gas leakage from the interior side of boss **116** (facing the interior environment **50** of the liner **120**) to the exterior side of boss **116** (facing the exterior environment **51** of the liner **120**). Moreover, any built-up gas between liner **120** and boss **116** can escape back into vessel **10** at port **32** of boss **116**, thereby preventing a rupture or separation at the interface of liner **120** and boss **116**. In an exemplary embodiment as shown in FIGS. **8** and **9**, pressure vessel **110** includes filament-wound outer shell **18** over liner **120** and exterior surface **62** of flange **130** of boss **116**.

[0044] As shown in FIGS. **7-9**, in a second exemplary embodiment of a boss **116***b*, lip **58** of ridge **54** is positioned at a greater distance from bore **32** compared to the first exemplary embodiment of a boss **116***a* of FIGS. **4-6**. For example, dimension C, a radially extending distance (normal to axis **42** between lip **58** and the bore surface **32**), is greater than T; in this case, the area between lip **58** and a mandrel disposed in port **32** is not fully filled with liner material.

[0045] Another difference between boss **116***b* and boss **116***a* is that floor **64** is provided at about a 100 degree obtuse angle alpha relative to port surface **32**, rather than a right angle as in boss **116***a*. This wider entry to the groove **44** may prevent voids from forming in the liner material as the liner **120** is roto-molded onto the interior side of boss **116***b*.

[0046] FIGS. **10** and **11** show cross-sectional partial slices of the sealing portion around ridge **54** of a third exemplary boss **116***c*, which is in some respects similar to boss **116***b* described above. In boss **116***c*, floor **64** has a radially inner portion **136** that is at a right angle to port surface **32** and a radially outer portion **138** that is provided at about a 170 degree obtuse angle relative to radially inner portion **136**. This wider entry to the groove **44***c* may prevent voids from forming in the liner material as the liner **120** is roto-molded onto the interior side of boss **116***c*.

[0047] Referring to FIG. **11**, the curvatures of portions of the ridge **54** and groove **46** are not uniform. For example, as interior surface **60** of flange **130** approaches the valley of groove **46** (from left to right), the extent of curvature of that surface increases (that is, the theoretical radius of the curvature is smaller or tighter) until an inflection point **132** between the concave surface of groove **46** and the convex surface of lip **52**. On the convex surface of lip **52**, the theoretical radius increases (the extent of curvature decreases—the curvature becomes more gradual) around the lip **52** from the inflection point **132** to a second inflection point **134** between the lip **52** and the straight portion of interior surface **56**.

[0048] On FIG. **11**, exemplary changes in effective curvature are illustrated in 45 degree arc segments. Beginning at the inflection point **134** (from right to left) on interior surface **56** for discussion, coming upward around the lip **52** in a counter-clockwise (leftward) direction, the first convex arc segment “I” may have a relatively gentle curvature described as having a theoretical radius of about 0.131 units (such as inches, for example). The next convex arc segment “II” may have a relatively smaller or tighter curvature described as having a theoretical radius of about 0.087 units. The next convex arc segment “III” may have a relatively smaller or tighter curvature described as having a theoretical radius of about 0.062 units. The next convex arc segment “IV” may have a relatively smaller or tighter curvature described as having a theoretical radius of about 0.044 units.

[0049] In an exemplary embodiment, a relatively straight segment “V” contains the inflection point **132**, and the groove **46** to the left of the segment “V” as illustrated has concave curvatures that are smallest or tightest near the straight segment “VI” and are more gradual extending out to the interior surface **60** of flange **130**. In an exemplary embodiment, there is no undercut (no rightward depression at segments “V” or “VI”) to prevent a void that may not fill with liner material.

[0050] Inflection point **132** marks the transition between the convex curves of arc segments “I-IV” and the concave curves of arc segments “VI-IX.” Each of the arc segments “VI-IX” also covers 45 degrees, though only the center of each of the arc segments “VI-IX” is indicated with an arrow for simplicity. Below inflection point **132**, the first concave arc segment “VI” may have a relatively tight curvature described as having a theoretical radius of about 0.066 units (such as inches, for example). The next concave arc segment “VII” may have a relatively larger or looser curvature described as having a theoretical radius of about 0.079 units. The next concave arc segment “VIII” may have a relatively larger or looser curvature described as having a theoretical radius of about 0.097 units. The next concave arc segment “IX” may have a relatively larger or looser curvature described as having a theoretical radius of about 1.306 units.

[0051] The described contours are very similar to a theoretical golden spiral, which is a logarithmic spiral that gets wider by a factor of the golden ratio for every quarter turn it takes. The golden ratio is a mathematical expression that is based on the Fibonacci sequence, where each term is the sum of the previous two terms. The golden ratio is approximately 1.618 to 1.

[0052] FIG. **15** is a partial cross-sectional view of a roto-molded liner **120** and boss **116**. As liner **120** fully cures, radial shrinkage of the liner occurs, designated by arrows **66**. This results in elevated contact pressure between liner **120** and boss **116** at the top of lip **52** as illustrated. This tight contact enhances sealing between liner **120** and the interior surfaces **56**, **60** of the boss **116**. Thus, high pressure sealing capability is achieved with the structure of the described boss **116** without the need for an auxiliary seal such as an O-ring or the complex two-part bosses often used with such O-rings. It has been found that the geometry of the substantially logarithmic structure of boss **116***c* FIGS. **10** and **11** may lead to desirably greater contact pressure in the primary seal area around lip **52** than the constant radius structures of bosses **116***a* and **116***b*.

[0053] Exemplary, non-limiting embodiments of a boss **116** and an assembly are described. In one aspect, a boss **116** is configured for attachment to a liner **120** of a pressure vessel **110**. The boss comprises a port **32**, a neck **28** and a flange **130**. The port **32** has a longitudinal axis **42** and is configured to connect an interior **50** of the pressure vessel and an exterior **51** of the pressure vessel **110**. The neck **28** circumscribes an exterior portion of the port **32**. The flange **130** circumscribes an interior portion of the port **32** and extends radially outward from the neck **28**. The flange **130** comprises an outer edge **68** at a farthest radial extent from the longitudinal axis **42**; an exterior side disposed on a first side of the outer edge **68**; and an interior side disposed on an opposed second side of the outer edge **68**. The interior side comprises a first radially inner annular groove **44** and a second radially outer annular groove **46** positioned at a greater distance from the longitudinal axis **42** than the first radially inner annular groove **44**. The first radially inner annular groove **44** has a first width D at its first base and a second width E at its first opening, wherein the second width E is equal to or greater than the first width D. The second radially outer annular groove **46** has a third width D at its second base and a fourth width E at its second opening, wherein the fourth width E is equal to or greater than the third width D.

[0054] In an exemplary embodiment, the interior side of the flange **130** comprises a ridge **54**. In an exemplary embodiment, the ridge **54** comprises a first interior surface **56** of the boss **116**; a first wall **140** of the first radially inner annular groove **44**; a first radially inner lip **58** connecting the first interior surface **56** of the boss and the first wall **140**; a second wall **142** of the second radially outer annular groove **46**; and a second radially outer lip **52** connecting the first interior surface **56** of the boss **116** and the second wall **142**. In an exemplary embodiment, the first interior surface **56** of the boss **116** is substantially perpendicular to the longitudinal axis **42**. In an exemplary embodiment, the interior side of the flange comprises a plurality of apertures **48** that extend between the first interior surface **56** of the boss **116** and the first wall **140**. In an exemplary embodiment, a radial dimension A of at least one of the plurality of apertures **48** is less than or equal to about twice the first width.

[0055] In an exemplary embodiment, a second interior surface **60** extends between the outer edge **68** and the second base of groove **46**, wherein the second base has a concave curvature. Referring to FIG. **11**, in an exemplary embodiment, a first radius of curvature of a first arc section (any one of VI, VII, VIII or IX) of the concave curvature is different from a second radius of curvature of a second arc section of the concave curvature (any other one of VI, VII, VIII or IX). In an exemplary embodiment, the first arc section is proximate the second wall **142**; the second arc section is proximate the second interior surface **60**; and the first radius of curvature is smaller than the second radius of curvature. In an exemplary embodiment, the concave curvature is substantially logarithmic.

[0056] In an exemplary embodiment, the second radially outer lip **52** has a convex curvature. In an exemplary embodiment, a first radius of curvature of a first arc section (any one of I, II, III or IV) of the convex curvature is different from a second radius of curvature of a second arc section of the convex curvature (any other one of I, II, III or IV). In an exemplary embodiment, the first arc section is proximate the first interior surface **56**; the second arc section is proximate the second wall **142**; and the first radius of curvature is greater than the second radius of curvature. In an exemplary embodiment, the convex curvature is substantially logarithmic.

[0057] In an exemplary embodiment, a width dimension of the first radially inner annular groove remains constant or increases continuously from the first width D at its first base to the second width E at its first opening; the width dimension of the first radially inner annular groove does not decrease as one measures from the first base toward the first opening.

[0058] In an exemplary embodiment, an assembly comprises a liner **120** comprising a first opening and first boss **116** attached to liner **120** at the first opening (see FIG. **9**). In an exemplary embodiment, a composite shell **18** is disposed on the liner **120** and on the exterior side of the flange **130**. In an exemplary embodiment, the liner **120** comprises a second opening, and the assembly comprises a second boss **116** attached to the liner **120** at the second opening (see FIG. **8**). In an exemplary embodiment, the liner **120** is formed by rotational molding. In an exemplary embodiment, the liner **120** has a substantially uniform thickness. In an exemplary embodiment, the liner **120** comprises an anchor **49** disposed proximate the first radially inner annular groove **44**, wherein the anchor **49** extends through an aperture **48** on an interior side of the flange **130**.

[0059] Although the subject of this disclosure has been described with reference to several embodiments, workers skilled in the art will recognize that changes may be made in form and detail without departing from the scope of the disclosure. All features described with reference to one embodiment are also applicable to other embodiments unless otherwise stated.

Claims

1. A boss configured for attachment to a liner of a pressure vessel, the boss comprising: a port having a longitudinal axis, the port configured to connect an interior of the pressure vessel and an exterior of the pressure vessel; a neck that circumscribes an exterior portion of the port; and a flange that circumscribes an interior portion of the port and that extends radially outward from the neck, wherein the flange comprises: an outer edge at a farthest radial extent from the longitudinal axis; an exterior side disposed on a first side of the outer edge; and an interior side disposed on an opposed second side of the outer edge, the interior side comprising: a first radially inner annular groove having a first width at its first base and a second width at its first opening, wherein the second width is equal to or greater than the first width; and a second radially outer annular groove positioned at a greater distance from the longitudinal axis than the first radially inner annular groove, the second radially outer annular groove having a third width at its second base and a fourth width at its second opening, wherein the fourth width is equal to or greater than the third width.

2. The boss of claim 1, wherein the interior side of the flange comprises a ridge comprising: a first interior surface of the boss; a first wall of the first radially inner annular groove; a first radially inner lip connecting the first interior surface of the boss and the first wall; a second wall of the second radially outer annular groove; and a second radially outer lip connecting the first interior surface of the boss and the second wall.

3. The boss of claim 2, wherein the first interior surface of the boss is substantially perpendicular to the longitudinal axis.

4. The boss of claim 2, wherein the interior side of the flange comprises a plurality of apertures that extend between the first interior surface of the boss and the first wall.

5. The boss of claim 4, wherein a radial dimension of at least one of the plurality of apertures is less than or equal to about twice the first width.

6. The boss of claim 2, comprising a second interior surface that extends between the outer edge and the second base, wherein the second base has a concave curvature.

7. The boss of claim 6, wherein a first radius of curvature of a first arc section of the concave curvature is different from a second radius of curvature of a second arc section of the concave curvature.

8. The boss of claim 7, wherein: the first arc section is proximate the second wall; the second arc section is proximate the second interior surface; and the first radius of curvature is smaller than the second radius of curvature.

9. The boss of claim 6, wherein the concave curvature is substantially logarithmic.

10. The boss of claim 2, wherein the second radially outer lip has a convex curvature.

11. The boss of claim 10, wherein a first radius of curvature of a first arc section of the convex curvature is different from a second radius of curvature of a second arc section of the convex curvature.

12. The boss of claim 11, wherein: the first arc section is proximate the first interior surface; the second arc section is proximate the second wall; and the first radius of curvature is greater than the second radius of curvature.

13. The boss of claim 10, wherein the convex curvature is substantially logarithmic.

14. The boss of claim 10, wherein a width dimension of the first radially inner annular groove remains constant or increases continuously from the first width at its first base to the second width at its first opening.

15. An assembly comprising: a liner comprising a first opening; and a first boss attached to the liner at the first opening, the first boss comprising: a port having a longitudinal axis, the port configured to connect an interior of a pressure vessel and an exterior of the pressure vessel; a neck that circumscribes an exterior portion of the port; and a flange that circumscribes an interior portion of the port and that extends radially outward from the neck, wherein the flange comprises: an outer edge at a farthest radial extent from the longitudinal axis; an exterior side disposed on a first side of the outer edge; and an interior side disposed on an opposed second side of the outer edge, the interior side comprising: a first radially inner annular groove having a first width at its first base and a second width at its first opening, wherein the second width is equal to or greater than the first width; and a second radially outer annular groove positioned at a greater distance from the longitudinal axis than the first radially inner annular groove, the second radially outer annular groove having a third width at its second base and a fourth width at its second opening, wherein the fourth width is equal to or greater than the third width.

16. The assembly of claim 15 comprising a composite shell disposed on the liner and on the exterior side of the flange.

17. The assembly of claim 15, wherein the liner comprises a second opening, the assembly comprising a second boss attached to the liner at the second opening.

18. The assembly of claim 15, wherein the liner is formed by rotational molding.

19. The assembly of claim 15, wherein the liner has a substantially uniform thickness.

20. The assembly of claim 15, wherein the liner comprises: an anchor disposed proximate the first radially inner annular groove; wherein the anchor extends through an aperture on an interior side of the flange.